NAJEEB ABDI

Computer Science Student | Applied AI Engineer

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EDUCATION

B.S. in Computer Science — *Expected May 2027*
Qatar University | Focus: Applied AI & Machine Learning — *Doha, Qatar*

EXPERIENCE

Industry Partnership Intern, Scale AI — *Jun 2026 – Aug 2026*
Qatar University Industry Partnership Program — *Doha, Qatar*

- Engineered the CS Intake Planner end to end: an agent-based Monte Carlo simulation of QU CS students moving through the real 41-course study plan (prerequisites, seat limits, pass/fail, dropout), built in Python and FastAPI.
- Turned the simulation into a decision tool that recommends the largest intake within department-set limits, with 95% uncertainty bounds, delivered as a Next.js dashboard for the Head of Department.
- Added a local-LLM advisor (Ollama) that explains results in plain language.

AI and Automation Intern — *Oct 2025 – Jan 2026*
Silatha (Remote) — *Amsterdam, Netherlands*

- Designed and deployed 3 production AI automation workflows (n8n, Python, OpenAI API) with third-party API integrations, monitoring, and error handling, replacing manual processes.
- Built and deployed a 6-level educational word search game (React, Firebase) with cross-device progress sync, played by ~200 players.

Research Intern, AI-Based Aquaponics System Development — *Aug 2025 – Nov 2025*
Qatar University — *Doha, Qatar*

- Joined a multi-institution AI research team building a federated aquaponics control system; set up the reinforcement learning model for distributed optimization.

PROJECTS

Thermal Trace, Co-Founder & Lead Developer | Open Innovation Award, UN Reboot the Earth Hackathon — *Sep 2026 – Present*
Cold-chain decision platform for Qatar/GCC (FastAPI, MQTT, PostgreSQL, Redis, React) — *Doha, Qatar*

- Led development in a 5-person team; built the Arrhenius shelf-life model that turns temperature history into remaining safe life.
- Enforced a "deterministic math owns every number" architecture: every value carries provenance, and the LLM only explains.
- Designed the end-to-end system: simulated refrigerated trucks and a real sensor node stream telemetry over MQTT into a FastAPI backend, where ML estimates spoilage risk, an optimizer picks the best cold store, and a food-loss engine reports kg and QAR saved.

Lectura, Solo Developer — *Mar 2026 – Oct 2026*
AI study tool that turns lecture videos into timestamped notes (FastAPI, faster-whisper, Next.js, Supabase, Docker) — *Doha, Qatar*

- Built an AI pipeline that turns lecture videos into timestamped notes: yt-dlp download, local faster-whisper transcription, chunking, and map-reduce LLM summarization with provider fallback across Groq, OpenAI, and local Ollama, plus optional slide OCR merged into the notes.
- Wrapped it in an authenticated FastAPI service (Supabase auth, row-level security, free-tier quotas of 3 videos/month, live progress streaming) with a Next.js frontend (seekable timestamps, chapters, export, "chat with your notes"), containerized with Docker; tested end to end on 5 lecture videos from 3 to 30 minutes, all processed successfully.

RouteyAI, Founder & Lead Developer — *Mar 2026 – Sep 2026*
School bus route planning and live tracking for Qatar schools — *Doha, Qatar*

- Built solo: K-means clustering plus nearest-neighbour and cheapest-insertion routing on Mapbox driving times, and web and mobile apps for admins, drivers, and parents (Next.js, Expo, Supabase/PostGIS).
- Wound down in Sep 2026 after the Ministry of Education rolled out its own national attendance system, which would have left the product with a single feature.

Taqriban, Solo Developer — *Oct 2026 – Present*
Arabic-first daily estimation game, live at taqriban.vercel.app — *Doha, Qatar*

- Built and launched a mobile-first, right-to-left game (Next.js, Supabase); all data access goes through Postgres security-definer functions with RLS deny-all, so answers stay server-side until a guess is locked.
- Set up CI (Vitest, Playwright, migration guard) and nightly backups; wrote 40 days of puzzles.

Other Projects: Naja7 AI (AIX Hackathon 2025), bilingual Arabic/English contract analysis prototype (Streamlit, OpenAI, Azure Translator).

ACTIVITIES & LEADERSHIP

Member, Events Committee, Artificial Intelligence Organization — *Apr 2025 – Present*
Qatar University — *Doha, Qatar*

- Help organize campus-wide AI events, including AI Week and the Innovation Program, coordinating logistics and demo setup.

Participant and Ambassador, Shell Intilaaqah Entrepreneurship Program — *Sep 2025 – Present*
Shell Qatar — *Doha, Qatar*

- Pitched an early-stage startup concept under mentor guidance; hosted (MC) the Fall 2026 masterclass "AI-Powered Innovation".

Also: Member, Events Team and Tech Team, ACM QU; Events Member, Mechatronics Committee; Student Assistant, IT Student Task Force.

SKILLS

AI / ML: Machine Learning, Deep Learning (TensorFlow), Reinforcement Learning, Federated Learning, Monte Carlo Simulation, LLM Apps (OpenAI, Groq, Ollama), Whisper / OCR, n8n
Web & Full-Stack: React, Next.js, TypeScript, Vite, Astro, Expo, FastAPI, Flask, REST APIs, WebSockets, Tailwind CSS
Data & Infrastructure: Supabase (PostgreSQL, PostGIS, Realtime, RLS), Firebase, Redis, MQTT, Mapbox GL JS
Testing & DevOps: pytest, Vitest, Playwright, CI/CD (GitHub Actions), Docker, Docker Compose, Vercel
Languages & Tools: Python, JS/TS, SQL, Java, HTML/CSS, Git, Postman | **Spoken:** English (Fluent), Arabic